

SURFACES AND 4-MANIFOLDS

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ABSTRACT. IAS/PCMI lecture notes.

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¹Based on lectures by David T. Gabai [10].

Part 1. Diffeomorphisms of 4-Manifolds²

1. DEHN TWISTS AND TWISTS ALONG SUBMANIFOLDS

All manifolds are smooth, compact, oriented, and fixed on the boundary.

1.1. Diffeomorphism groups, Lefschetz singularities, and Dehn twists.

Definition 1.1.1 (Mapping classes and mapping tori). For $\phi \in \text{Diff}(X)$,

$$\pi_0(\text{Diff}(X)) = \text{Diff}(X)/\text{isotopy}, \quad Y_\phi = [0, 1] \times X / (1, x) \sim (0, \phi(x)).$$

A family $[0, 1]^k \rightarrow \text{Diff}(X)$ means a smooth map $[0, 1]^k \times X \rightarrow X$, and $Y_\phi \rightarrow S^1$ has monodromy ϕ .

Remark 1.1.2 (Homotopy-type reconstruction). Abstract group reconstruction holds [4], whereas

$$\text{Diff}(X) \simeq \text{Diff}(X') \implies X \cong X'$$

for closed surfaces of distinct genera at least two.

Fact 1.1.3 (Ball diffeomorphisms).

n	$\text{Diff}_\partial(B^n)$
1, 2, 3	$\simeq \{\text{id}\}$
4	π_0 unknown, $\pi_k \neq 0$ for some $k \geq 1$
5	$\pi_0 = 0$
6	$\pi_0 \neq 0$

The first two rows use [19, 13, 20].

Proposition 1.1.4 (Alexander trick, 3.1).

$$\pi_0(\text{Homeo}_\partial(B^n)) = 0.$$

Proof. Radial contraction gives the isotopy.

$$H_t(x) = \begin{cases} t h(x/t), & |x| \leq t, \\ x, & t \leq |x| \leq 1, \end{cases} \quad H_0 = \text{id}, \quad H_1 = h,$$

$$h|_{S^{n-1}} = \text{id}, \quad \sup_{x \in B^n} |H_t(x) - x| \leq 2t \implies H \in C^0([0, 1] \times B^n, B^n).$$

Thus $[h] = [\text{id}]$. □

Proposition 1.1.5 (The interval, 3.2).

$$\pi_0(\text{Diff}_\partial(B^1)) = 0.$$

Proof. Convex interpolation works in dimension one.

$$f_t(x) = (1-t)x + tf(x), \quad f'_t(x) = (1-t) + tf'(x) > 0, \quad f_t(\pm 1) = \pm 1,$$

$$f(r, \theta) = (r, \theta + \pi \eta(r)), \quad D\left(\frac{\text{id} + f}{2}\right)_0 = 0.$$

Thus the first path is an isotopy, while its higher-dimensional analogue may be singular. □

Proposition 1.1.6 (Balls and spheres, 3.3).

$$\pi_0(\text{Diff}_\partial(B^n)) \cong \pi_0(\text{Diff}(S^n)).$$

²Based on lectures by David T. Gabai [10].

Proof. The oriented first jet gives a split fibration [18].

$$\begin{aligned} \text{Diff}_\partial(B^n) &\simeq F_q \longrightarrow \text{Diff}(S^n) \xrightarrow{j_q^1} \text{SO}(n+1), & j_q^1 \circ s &= \text{id}, \\ \pi_1(\text{Diff}(S^n)) &\longrightarrow \pi_1(\text{SO}(n+1)) \xrightarrow{\partial} \pi_0(F_q) \longrightarrow \pi_0(\text{Diff}(S^n)) \longrightarrow 0, \\ (j_q^1)_* \text{ surjective} &\implies \partial = 0 \implies \pi_0(F_q) \cong \pi_0(\text{Diff}(S^n)). \end{aligned}$$

Thus the mapping class groups agree. $\square$

Proposition 1.1.7 (Commutativity, 3.4). *The groups in Proposition 1.1.6 are abelian.*

Proof. Conjugate representatives into disjoint balls.

$$\begin{aligned} [\bar{f}] &= [f], & [\bar{g}] &= [g], & \text{supp}(\bar{f}) \cap \text{supp}(\bar{g}) &= \emptyset, \\ \bar{f}\bar{g} &= \bar{g}\bar{f} \implies [f][g] = [g][f] \xrightarrow{1.1.6} \pi_0(\text{Diff}(S^n)) \text{ abelian.} \end{aligned}$$

Thus both groups commute. $\square$

Proposition 1.1.8 (Mapping-torus detection, 3.5). *A bundle isomorphism $Y_\phi \cong Y_{\phi'}$ gives*

$$[\phi'] = [h\phi h^{-1}],$$

and a fiber-fixing isomorphism gives $[\phi'] = [\phi]$.

Proof. Cut the base at one point.

$$\begin{aligned} F([t, x]) &= [t, h_t(x)], & h_0\phi &= \phi'h_1, & [h_0] &= [h_1], \\ [\phi'] &= [h_0][\phi][h_0]^{-1}, & F|_{\{0\} \times X} &= \text{id} \implies h_0 = \text{id} \implies [\phi'] = [\phi]. \end{aligned}$$

Thus the bundle determines the conjugacy class of its monodromy. $\square$

Proposition 1.1.9 (Singular extension, 3.6). *Every smooth $f: \partial Z \rightarrow S^1$ extends to $F: Z \rightarrow D^2$, generally with codimension-zero critical set.*

Proof. Use a collar $c(y, s)$ and a flat cutoff χ .

$$\begin{aligned} F(c(y, s)) &= \chi(s)f(y), & F|_{Z \setminus c(\partial Z \times [0, \varepsilon])} &= 0, \\ \chi^{(k)}(\varepsilon) &= 0 \quad (k \geq 0) \implies F \in C^\infty(Z, D^2), & Z \setminus c(\partial Z \times [0, \varepsilon]) &\subseteq \text{Crit}(F). \end{aligned}$$

Thus the extension exists with a large singular locus. $\square$

Proposition 1.1.10 (Nonsingular filling, 3.7). *If $Y_\phi \rightarrow S^1$ extends to a submersion $Z \rightarrow D^2$, then $\phi \simeq \text{id}$.*

Proof. Ehresmann trivializes the filling [3].

$$\begin{aligned} \text{Crit}(p) &= \emptyset \implies (Z, p) \cong (D^2 \times X, \text{pr}_{D^2}), \\ Y_\phi &\cong S^1 \times X = Y_{\text{id}} \xrightarrow{1.1.8} \phi \simeq \text{id}. \end{aligned}$$

Thus nontrivial monodromy requires singularities. $\square$

Proposition 1.1.11 (Quadratic critical point, 3.8). *For $p(z_0, \dots, z_m) = \sum_{j=0}^m z_j^2$,*

$$\begin{aligned} Dp_{\mathbf{z}}(\mathbf{v}) &= 2 \sum_{j=0}^m z_j v_j, & \text{Crit}(p) &= \{0\}, \\ \text{rank}_{\mathbb{R}}(Dp_0) &= 0, & \text{rank}_{\mathbb{R}}(Dp_{\mathbf{z}}) &= 2 \quad (\mathbf{z} \neq 0). \end{aligned}$$

Definition 1.1.12 (Model and ambient Dehn twists). For

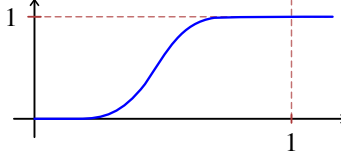
$$TS^m = \{(\mathbf{x}, \mathbf{y}) : |\mathbf{x}| = 1, \mathbf{x} \cdot \mathbf{y} = 0\},$$

set

$$R_\rho(\mathbf{x}, \mathbf{y}) = \left(\cos \rho \mathbf{x} + \sin \rho \frac{\mathbf{y}}{|\mathbf{y}|}, -\sin \rho |\mathbf{y}| \mathbf{x} + \cos \rho \mathbf{y} \right).$$

Choose $\tau = 0$ near 0, $\tau = 1$ near $[1, \infty)$, let $a(\mathbf{x}, \mathbf{y}) = (-\mathbf{x}, -\mathbf{y})$, and define

$$\delta = a \circ R_{\pi\tau(|\mathbf{y}|)}, \quad \delta_{(c,v)} = \begin{cases} v\delta v^{-1}, & \text{on } v(DTS^m), \\ \text{id}, & \text{outside } v(DTS^m). \end{cases}$$



Theorem 1.1.13 (Lefschetz monodromy, 3.9). For $p(\mathbf{z}) = \sum z_j^2$,

(1)

$$p(\mathbf{x} + i\mathbf{y}) = r \iff |\mathbf{x}|^2 = r + |\mathbf{y}|^2, \quad \mathbf{x} \cdot \mathbf{y} = 0.$$

(2)

$$\phi_r(\mathbf{x}, \mathbf{y}) = \sqrt{r + |\mathbf{y}|^2} \mathbf{x} + i\mathbf{y}, \quad \phi_r : TS^m \xrightarrow{\cong} p^{-1}(r).$$

(3)

$$F_{r,\theta} = e^{i\theta/2} \phi_r, \quad F_{r,\theta} : TS^m \xrightarrow{\cong} p^{-1}(re^{i\theta}).$$

(4)

$$R_\rho R_\sigma = R_{\rho+\sigma}, \quad R_\pi = a, \quad \psi_{r,\theta} = e^{i\theta/2} \phi_r R_{(\theta/2)\tau}.$$

(5)

$$\psi_{r,2\pi} = \psi_{r,0} \circ a R_{\pi\tau} = \psi_{r,0} \circ \delta.$$

(6)

$$p^{-1}(S^1) \cong TS^m \times [0, 2\pi] / (u, 2\pi) \sim (\delta(u), 0).$$

Hence a Lefschetz point has monodromy δ along its vanishing sphere.

Proof. The half-angle trivialization closes by the model twist.

$$p(e^{i\theta/2} \phi_r(u)) = re^{i\theta}, \quad \phi_r(-u) = -\phi_r(u), \quad R_{\pi\tau} = a\delta,$$

$$\psi_{r,2\pi}(u) = -\phi_r(R_{\pi\tau}u) = \phi_r(\delta u) = \psi_{r,0}(\delta u).$$

Thus the clutching map is δ . □

Remark 1.1.14 (Rounded Milnor tube).

$$M_{R,\varepsilon} = \{\mathbf{z} : |\mathbf{z}| \leq R, |p(\mathbf{z})| \leq \varepsilon\}, \quad 0 < \varepsilon \ll R^2.$$

After rounding corners, Ehresmann trivializes its radial boundary [3].

Proposition 1.1.15 (Isotopy invariance, 3.10). *Transported tubular data give isotopic twists.*

Proof. Conjugation by isotopy extension suffices.

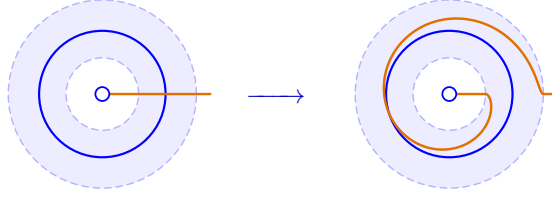
$$C_t = H_t(C_0), \quad \mathbf{v}_t = H_t \mathbf{v}_0, \quad \delta_{(c_t, \mathbf{v}_t)} = H_t \delta_{(c_0, \mathbf{v}_0)} H_t^{-1}.$$

Thus the twist depends only on the isotopy class of its tubular data. □

Proposition 1.1.16 (The annular twist, 3.11). *For $m = 1$, δ is the right-handed annular Dehn twist.*

Proof. The model is one positive winding.

$$\delta(q, s) = (q + \alpha(s), s), \quad \alpha(-1) = 0, \quad \alpha(1) = 2\pi, \quad \alpha'(s) \geq 0.$$



Thus the relative winding is +1. □

Proposition 1.1.17 (Local nontriviality, 3.12).

$$[\delta] \neq 1 \in \pi_0(\text{Diff}_\partial(DTS^m)).$$

Proof. The available detectors are

$$\begin{aligned} m = 1 &\implies w(\delta\gamma) - w(\gamma) = 1, & m \text{ even} &\implies \delta_* = -\text{id on } H_m(DTS^m; \mathbb{Z}), \\ m \geq 3 \text{ odd} &\xRightarrow{[15, \text{Theorem 1.1}]} \text{ord}([\delta]) = \infty. \end{aligned}$$

Thus $[\delta] \neq 1$ for every $m \geq 1$. □

Proposition 1.1.18 (The four-dimensional twist, 3.13). *For $m = 2$, $[\delta]$ has order two.*

Proof. The square untwists, while the zero section detects δ .

$$\begin{aligned} \delta^2 &= R_{2\pi\tau} \xRightarrow{[15, \text{Proposition 3.1}]} \delta^2 \simeq \text{id} \quad \text{rel. } \partial DTS^2, \\ \delta|_{S^2} &= a, \quad \deg(a) = -1 \implies \delta_* = -\text{id on } H_2(DTS^2; \mathbb{Z}). \end{aligned}$$

Thus $\text{ord}([\delta]) = 2$. □

Fact 1.1.19 (Higher-dimensional orders).

$$\text{ord}([\delta]) = \begin{cases} \infty, & m \text{ odd}, \\ 2, & m = 2, 6, \\ 4, & m \text{ even and the Kervaire sphere is standard}, \\ 8, & m \text{ even and the Kervaire sphere is exotic} \end{cases}$$

by [15, Theorem 1.1].

1.2. Codimension-one twists, Montesinos twins, and barbells. For $C^{n-1} \subset X^n$ and $\psi_t \in \pi_1(\text{Diff}(C))$, define

$$\tau_C(t, p) = (t, \psi_t(p)) \quad \text{on } [0, 1] \times C, \quad \tau_C = \text{id} \quad \text{elsewhere.}$$

Proposition 1.2.1 (Bounding three-spheres, 3.14). *If $S^3 = \partial B^4 \subset X$, then*

$$\tau_{S^3} \simeq \text{id}.$$

Proof. Absorb the rotational loop $A_u \in \text{SO}(4)$ at the center.

$$h_s(r\mathbf{v}) = rA_{s+(1-s)\rho(r)}\mathbf{v}, \quad h_0 = \tau_{S^3}, \quad h_1 = \text{id}, \quad h_s|_{\partial B^4} = \text{id}.$$

Thus extension by the identity gives the ambient isotopy. □

Proposition 1.2.2 (Bounding product twists, 3.15). *If*

$$S^1 \times \Sigma = \partial(D^2 \times \Sigma) \subset X,$$

then its product twist is isotopic to the identity.

Proof. Use the disk-factor isotopy.

$$H_s(r\mathbf{v}, q) = (rA_{s+(1-s)\rho(r)}\mathbf{v}, q) = h_s \times \text{id}_\Sigma,$$

$$H_0 = \tau_{S^1 \times \Sigma}, \quad H_1 = \text{id}.$$

Thus the product twist is trivial. □

Proposition 1.2.3 (The rotation loop). *For*

$$r_t(z, x) = (e^{2\pi i t} z, x),$$

one has

$$(\text{pr}_{S^1} \circ \text{ev}_{(1,q)})_*[r] = [e^{2\pi i t}] = 1 \in \pi_1(S^1) \implies [r] \neq 0 \in \pi_1(\text{Diff}(S^1 \times \Sigma)).$$

Definition 1.2.4 (Framed Montesinos twins and their twists). A Montesinos twin $W = (R, S) \subset X^4$ consists of square-zero spheres meeting twice with opposite signs [9]. A framing chooses

$$\partial \mathbf{v}(R \cup S) \cong S_\lambda^1 \times S_R^1 \times S_S^1.$$

The twist τ_W is the Dehn twist on $[0, 1] \times S_\lambda^1$, crossed with $\text{id}_{S_R^1 \times S_S^1}$.

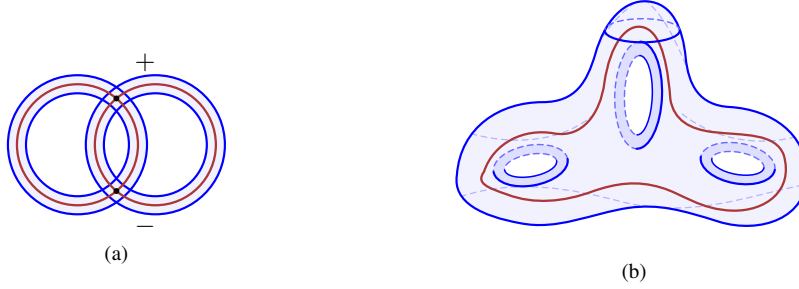


FIGURE 1. A half-dimensional schematic of a Montesinos twin and its embedded regular neighborhood $\Sigma_{0,4}$.

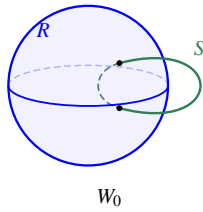


FIGURE 2. An $\mathbb{R}^3$ -slice of the trivial twin $W_0 = (R, S)$.

Proposition 1.2.5 (Twin boundary, 3.16).

$$\partial \mathbf{v}(R \cup S) \cong S_\lambda^1 \times S_R^1 \times S_S^1,$$

where S_R^1, S_S^1 are meridians and S_λ^1 is the zero-framed longitude.

Proof. The two opposite plumbings cancel on the boundary.

$$\begin{array}{ccc} S_\lambda^1 \times B^2 \times S_S^1 & \xrightarrow{\text{surg}_{\{1\} \times \{0\} \times S_S^1}} & \nu(R \cup S) \\ \partial \downarrow & & \downarrow \partial \\ S_\lambda^1 \times S_R^1 \times S_S^1 & \xlongequal{\quad} & S_\lambda^1 \times S_R^1 \times S_S^1. \end{array}$$

Thus the factors are the longitude and two meridians. $\square$

Proposition 1.2.6 (Twin orientations, 3.17).

$$[\tau_{(R,S)}]^{-1} = [\tau_{(S,R)}] = [\tau_{(-R,S)}] = [\tau_{(R,-S)}].$$

Proof. Each operation reverses the longitude.

$$(R, S) \mapsto (S, R), (-R, S), (R, -S) \implies o_\lambda \mapsto -o_\lambda \implies [D_\lambda] \mapsto [D_\lambda]^{-1}.$$

Thus every displayed change inverts the twist [9, Lemma 3]. $\square$

Proposition 1.2.7 (The trivial twin, 3.18). *For W_0 in Figure 2,*

$$S_\lambda^1 \times S_R^1 \times S_S^1 = \partial(D_\lambda^2 \times S_R^1 \times S_S^1) \xrightarrow{1.2.2} \tau_{W_0} \simeq \text{id}.$$

Fact 1.2.8 (Half-unknotted twists). *Half-unknotted Montesinos twists generate either 0 or $\mathbb{Z}/2\mathbb{Z}$, with generator τ_{W_*} [11, Theorem 4].*

Theorem 1.2.9 (Announced Gabai–Gay–Hartman theorem). *The announced sequel asserts*

$$[\tau_{W_*}] \neq 1 \in \pi_0(\text{Diff}(S^4))$$

[6, 7].

Proposition 1.2.10 (Surgery description, 3.19). *Let $X^* = \text{surg}_R(X)$.*

(1) *S becomes a torus $T \subset X^*$.*

(2)

$$T = \bigcup_{a \in S^1} \alpha_a(S^1), \quad \alpha_0(S^1) = \mu_R.$$

(3)

$$\text{surg}_{\mu_R}(X^*) = X.$$

(4) *Circle pushing along $\{\alpha_a\}$, followed by reverse surgery, equals τ_W .*

Proof. The surgeries and circle push form

$$\begin{aligned} (X; R, S) &\xrightarrow{\text{surg}_R} (X^*; \mu_R, T) \xrightarrow{\text{push}_{\{\alpha_a\}}} (X^*; \mu_R, T) \xrightarrow{\text{surg}_{\mu_R}} (X; R, S). \\ S \setminus (D_+ \sqcup D_-) &\cong [0, 1] \times S^1, \quad T \cong S^1 \times S^1, \\ [\text{surg}_{\mu_R} \text{push}_{\{\alpha_a\}}] &= [\tau_W] \end{aligned}$$

[9, 11]. Thus the trace torus recovers the twin twist. $\square$

Proposition 1.2.11 (Spun-knot twins, 3.20). *Every spun-knot twin has trivial Montesinos twist.*

Proof. After surgery on the axis, its trace loop contracts.

$$X^* \cong S^1 \times S^3, \quad \alpha_t(z) = (z, K(t)), \quad \pi_1(S^3) = 0,$$

$$\{\alpha_t\} \simeq \{S^1 \times \{p\}\} \implies \text{push}_{\{\alpha_t\}} \simeq \text{id} \xrightarrow{1.2.10} \tau_W \simeq \text{id}.$$

Thus the twist vanishes. $\square$

Proposition 1.2.12 (Barbell equivalence, 3.21). *For a barbell $R \cup A \cup S$, the finger-move twin twist and circle-pushing map agree.*

Proof. Surgery converts the finger trace into the pushing loop.

$$\begin{array}{ccccc}
 R & \xrightarrow{\text{finger}_{A\mu_S\bar{A}}} & R' & \xrightarrow{\text{surg}_R} & \{\alpha_t(R^*)\}_{t \in S^1} \\
 \downarrow \text{isotopy extension} & & & & \downarrow \text{framed isotopy extension} \\
 & & \tau_{(R',S)} & \xleftarrow{\text{surg}_{R^*}} & \text{push}_{A\mu_S\bar{A}}.
 \end{array}$$

Thus the constructions determine the same mapping class [9, §6]. $\square$

Definition 1.2.13 (Stable maps). A smooth map $f: A \rightarrow B$ is stable when every small family f_t satisfies

$$\psi_t^{-1} \circ f_t \circ \phi_t = f_0, \quad \phi_0 = \text{id}_A, \quad \psi_0 = \text{id}_B.$$

Fact 1.2.14 (Mather stability). *Stable proper maps are open and dense in the nice dimensions [17]. Stable maps to surfaces are Morse 2-functions [12].*

Proposition 1.2.15 (Rank-zero instability). *A germ*

$$(A^n, p) \longrightarrow (B^2, f(p)), \quad df_p = 0,$$

is unstable.

Proof. One-jet transversality removes the rank-zero stratum.

$$\begin{aligned}
 \Sigma^0 &= \{j_x^1 g : Dg_x = 0\}, \quad \text{codim}(\Sigma^0) = 2n > n, \\
 j^1 f_t \pitchfork \Sigma^0 &\implies (j^1 f_t)^{-1}(\Sigma^0) = \emptyset \implies f_t \not\sim f_0.
 \end{aligned}$$

Thus rank-zero critical points are unstable [17]. $\square$

Proposition 1.2.16 (The cubic, 3.22). *The map $f(x) = x^3$ is not stable.*

Proof. The perturbation deletes its critical point.

$$\begin{aligned}
 f_t(x) &= x^3 + tx\eta(x), \quad \eta = 1 \text{ near } 0, \quad 0 < t \ll 1, \\
 f'_t(x) &> 0, \quad \text{Crit}(f_t) = \emptyset, \quad \text{Crit}(f_0) = \{0\}.
 \end{aligned}$$

Thus $f_t \not\sim f_0$. $\square$

Proposition 1.2.17 (The quadratic, 3.23). *The map $f(x) = x^2$ is stable.*

Proof. Morse factorization supplies the coordinates.

$$\begin{aligned}
 f'_t(c_t) &= 0, \quad m_t = f_t(c_t), \quad f_t(x) - m_t = (x - c_t)^2 v_t(x), \quad v_t(x) > 0, \\
 u_t(x) &= (x - c_t)\sqrt{v_t(x)}, \quad f_t(x) = u_t(x)^2 + m_t, \\
 \phi_t &= u_t^{-1}, \quad \psi_t(y) = y + m_t, \quad \psi_t^{-1} f_t \phi_t(x) = x^2.
 \end{aligned}$$

Thus x^2 is stable. $\square$

2. CERF GRAPHICS, EYES, AND THE FINGER–WHITNEY CALCULUS

2.1. Morse 2-functions, Cerf graphics, and eye graphics.

Fact 2.1.1 (Morse 2-functions). *For closed oriented X^4 , $\phi \in \text{Diff}(X)$, $\partial Z^6 = Y_\phi$, and a stable extension $F: Z \rightarrow D^2$, the local singular models are*

$$(2.1.1) \quad (t, x) \mapsto (t, -x_1^2 - \cdots - x_k^2 + x_{k+1}^2 + \cdots + x_{n-1}^2),$$

$$(2.1.2) \quad (t, x) \mapsto (t, -x_1^2 - \cdots - x_k^2 + x_{k+1}^3 - tx_{k+1} + x_{k+2}^2 + \cdots + x_{n-1}^2).$$

The fold has transverse index k , or $n-1-k$ from the opposite coorientation, while the cusp creates or cancels indices $k, k+1$. These notes follow [10]; the calculus below follows [6], with stability supplied by Fact 1.2.14.

Definition 2.1.2 (Singular locus and graphic).

$$\text{Crit}(F) = \{z : \text{rank}(dF_z) < 2\}, \quad \Gamma(F) = F(\text{Crit}(F)).$$

These are the singular locus and singular graphic.

Lemma 2.1.3 (Embedded singular locus, 3.26, 3.24). *The singular locus is a smoothly embedded 1-manifold; its cusp image is not a smooth embedded curve.*

Proof. Differentiate (2.1.1)–(2.1.2).

$$\begin{aligned} \text{Crit}(F)_{\text{fold}} &= \{(t, 0, \dots, 0)\} \cong \mathbb{R}, \\ u = x_{k+1}, \quad \text{Crit}(F)_{\text{cusp}} &= \{x_i = 0 \ (i \neq k+1), \ t = 3u^2\}, \\ \gamma(u) &= (3u^2, 0, \dots, u, \dots, 0), \quad \gamma'(u) = (6u, 0, \dots, 1, \dots, 0) \neq 0, \\ F\gamma(u) &= (3u^2, -2u^3), \quad T_0\Gamma(F) = \mathbb{R}(1, 0), \\ 3u^2 = 3(-u)^2 \ (u \neq 0) &\implies \Gamma(F) \text{ is not smoothly embedded at } 0. \end{aligned}$$

Thus the domain locus is embedded and its cusp image is semicubical. □

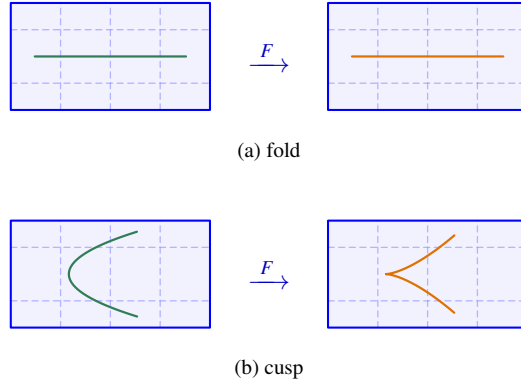


FIGURE 3. The smooth critical loci project to the fold and semicubical cusp graphics.

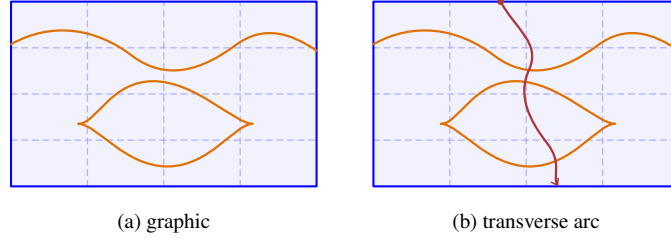


FIGURE 4. A Morse 2-function graphic with cooriented folds and a transverse oriented arc.

Definition 2.1.4 (Pseudoisotopy). A pseudoisotopy from id to ϕ is

$$\Phi \in \text{Diff}([0, 1] \times X), \quad \Phi(0, p) = (0, p), \quad \Phi(1, p) = (1, \phi(p)),$$

with $\Phi(t, p) = (t, p)$ for $p \in \partial X$.

Proposition 2.1.5 (Global time and pseudoisotopy, 3.25, 3.27, 3.28). *Let $h = \pi_t \circ F$, assume product boundary behavior; and suppose the graphic has no vertical tangencies.*

(1) *If h is submersive on every vertical-boundary stratum, then*

$$Z \cong [0, 1] \times Y, \quad Y = F^{-1}(\{0\} \times [0, 1]), \quad F(t, p) = (t, f_t(p)).$$

(2) *Under the analogous second-coordinate hypotheses,*

$$Y \cong [0, 1] \times X, \quad Z \cong [0, 1]^2 \times X,$$

and only finitely many times contain cusp births or deaths.

(3) *For a cut-open filling of $Y_\phi \rightarrow S^1$,*

$$\text{Crit}(f_0) = \text{Crit}(f_1) = \emptyset, \quad \phi \simeq_{\text{pseudo}} \text{id}.$$

Proof. The target coordinates and cut-face gluing give the result.

$$z \notin \text{Crit}(F) \implies \text{rank}(dF_z) = 2 \implies dh_z \neq 0, \quad z \in \text{Crit}(F) \implies \text{im}(dF_z) \not\subseteq \ker(d\pi_t),$$

$$h \text{ proper and stratumwise submersive} \xrightarrow{[31]} Z \cong [0, 1] \times Y \xrightarrow{[31]} Z \cong [0, 1]^2 \times X,$$

$$\hat{\phi}(y, p) = (y, \phi(p)), \quad f_1 \circ \hat{\phi} = f_0, \quad \text{Crit}(f_1) = \hat{\phi}(\text{Crit}(f_0)) = \emptyset,$$

$$\begin{array}{ccc} [0, 1] \times X & \xrightarrow{\tilde{\Xi}} & \times X \\ q_\phi \downarrow & & \downarrow q_{\text{id}} \\ Y_\phi & \xrightarrow{\Xi} & Y_{\text{id}} \end{array} \quad \begin{array}{l} \tilde{\Xi}(0, p) = (0, a(p)), \\ \tilde{\Xi}(1, p) = (1, a\phi(p)), \\ \Phi = (\text{id} \times a^{-1})\tilde{\Xi}. \end{array}$$

Thus horizontal time gives the product and Φ gives the pseudoisotopy. □

Fact 2.1.6 (Cerf and Kreck). *Level-preserving pseudoisotopy is isotopy, while a generic interpolation has graphic*

$$\Gamma = \{(t, z) : z \in \text{CritVal}(f_t)\}$$

[2]. *If X is closed and simply connected and $\phi_* = \text{id}$ on $H_*(X; \mathbb{Z})$, then*

$$\phi \simeq_{\text{pseudo}} \text{id}$$

[16, Theorem 1]; see [8, §1.3].

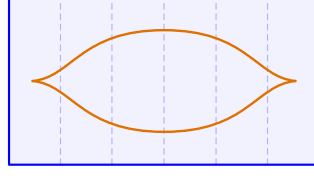


FIGURE 5. An eye graphic, with birth and death cusps joined by two folds.

Proposition 2.1.7 (Eye data, 3.29, 3.30, 3.31). *For a family with Morse points and finitely many births or deaths:*

- (1) *a smooth vertical family V_t , standard near every singularity, satisfies $df_t(V_t) > 0$ off $\text{Crit}(f_t)$;*
- (2) *below a $(k, k+1)$ -birth the descending trace is B^k , splitting after birth into S^{k-1} and $S^k \setminus \mathring{D}^k$;*
- (3) *if X is simply connected, a loop in $\text{Diff}(X)$ makes the S^0 -data constant throughout a $(1, 2)$ -eye and the B^1 -data constant near its cusps.*

Proof. The cusp chart, partition of unity [14], its lower level, and isotopy extension give the claims.

$$\begin{aligned}
 f_t &= -|x^-|^2 + u^3 - tu + |x^+|^2, & V_t &= -2x^- \frac{\partial}{\partial x^-} + (3u^2 - t) \frac{\partial}{\partial u} + 2x^+ \frac{\partial}{\partial x^+}, \\
 df_t(V_t) &= 4|x^-|^2 + (3u^2 - t)^2 + 4|x^+|^2 > 0, & V_t &= \sum_i \chi_i V_{i,t}, \\
 D_0^- \cap f_0^{-1}(-\delta) &= \{(x^-, u, 0) : |x^-|^2 + |u|^3 = \delta, u \leq 0\} \cong B^k, \\
 c_\pm &= (0, \pm \sqrt{t/3}, 0), & \text{ind}(c_+) &= k, \quad \text{ind}(c_-) = k+1, & B^k &\longleftrightarrow S^{k-1} \cup (S^k \setminus \mathring{D}^k), \\
 \begin{array}{ccc} B^1 & \xrightarrow{\beta_a} & X \\ \parallel & & \downarrow H_t \\ B^1 & \xrightarrow{\beta_t} & X \end{array} & \pi_1(X) = 0 \implies H_{b-\varepsilon} \beta_a \simeq_{\text{isot.}} \beta_{b-\varepsilon} \text{ rel. } S^0, \\
 H_0 = H_1 &= \text{id}, & \psi_t &= H_t^{-1}, & \psi_t \alpha_t &= \alpha_a, & \psi_t \beta_t &= \beta_a.
 \end{aligned}$$

Thus the eye admits the asserted gradient, descending, and straightened data. $\square$

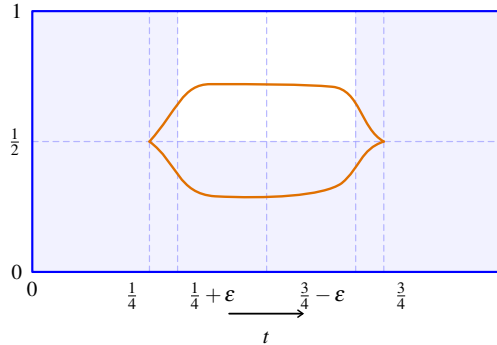


FIGURE 6. The standardized region; only the central top gap remains.

Warning 2.1.8 (Framings). Every attaching sphere carries its framed thickening; the pictures suppress this data.

Proposition 2.1.9 (Dotted 2-spheres, 3.32). *A dotted unknotted 2-sphere in X^4 records a 5-dimensional 1-handle on $X \times [0, 1]$.*

Proof. Read the outgoing boundary by dual surgery.

$$\begin{aligned} (M^n \times I) \cup_{S^0 \times D^n} (D^1 \times D^n) &\Rightarrow \partial_+ \cong M\#(S^1 \times S^{n-1}), \\ n = 3 : \quad (M \setminus S^1 \times \mathring{D}^2) \cup (D^2 \times S^1) &\cong M\#(S^1 \times S^2), \\ n = 4 : \quad (X \setminus S^2 \times \mathring{D}^2) \cup (D^3 \times S^1) &\cong X\#(S^1 \times S^3), \\ S^1 \subset M \longleftrightarrow S^2 \subset X, \quad \text{dot} &\longleftrightarrow \text{1-handle provenance.} \end{aligned}$$

Thus the dotted 2-sphere is the five-dimensional dotted-handle notation. $\square$

Proposition 2.1.10 (Circle pushing and Montesinos twists, 3.33). *Circle pushing is Montesinos twisting; the 1-2 subgroup is generated by twins having at least one unknotted sphere.*

Proof. Normalize a framed loop of circles and surger its trace torus.

$$\begin{aligned} \alpha : S^1_t &\rightarrow \text{Emb}^{\text{fr}}(S^1_z, S^1 \times S^3), \quad T(t, z) = \alpha_t(z), \quad C = S^1 \times \{p\}, \\ \nu T &= D^2_{r, \theta} \times S^1_a \times S^1_b, \quad P_\alpha(r, \theta, a, b) = (r, \theta, a + h(r), b), \\ (S^1 \times S^3, C \subset T) &\xrightarrow{\text{surge}_C} (S^4, W = (R, S)), \quad P_\alpha \mapsto \tau_W, \\ T \text{ surg } C &= R, \quad \text{surgery core} = S, \quad R \cap S = \{p_+, p_-\}, \quad \varepsilon(p_+) = -\varepsilon(p_-), \\ \text{im}(\text{Push}) &= \langle \tau_W : R \text{ or } S \text{ unknotted} \rangle \xrightarrow{[11, \text{Lemma 6}]} \text{1-2 subgroup.} \end{aligned}$$

Hence circle pushing and half-unknotted twin twisting agree [11, Corollary 7]. $\square$

Theorem 2.1.11 (Single-eye reformulation, [9, Theorem 18]). *Every pseudoisotopy of S^4 using only indices 1, 2 has a Cerf representative with one 1-2 eye, by the Hatcher–Wagoner reduction [14, Chapter VI, Propositions 3–4].*

Theorem 2.1.12 (Announced trisected extension, [10]). *It is announced that, if X is closed, simply connected and spin, every Y_ϕ bounds a compact simply connected spin Z^6 carrying an extending Morse 2-function whose inward-index-2 graphic braids about the center and meets each trisection sector in components with at most one cusp.*

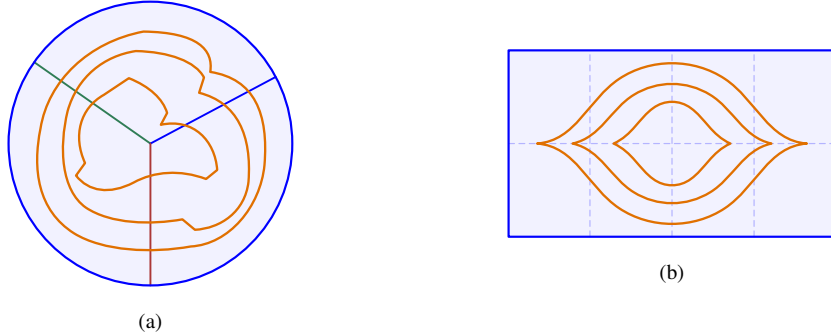


FIGURE 7. The trisected extension graphic and nested 2-3 eyes.

Theorem 2.1.13 (Nested 2-3 eyes, [14, Chapter VI, Proposition 3], [6, Introduction], [10]). *Every orientation-preserving $\phi \in \text{Diff}(S^4)$ has*

$$Y_\phi \cong \partial([0, 1]^2 \times S^4)$$

and an extending family

$$(t, y, x) \mapsto (t, f_t(y, x))$$

with a nested 2-3 eye graphic and no handle slides; moreover

$$\phi \simeq_{\text{pseudo}} \text{id}$$

[1, p. 157].

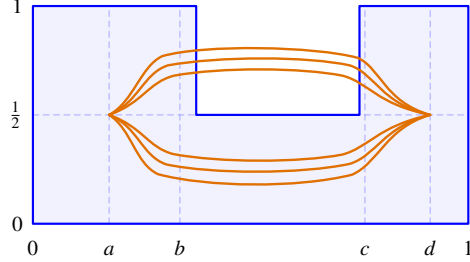


FIGURE 8. The standardized nested-eye region.

Fact 2.1.14 (Nested-eye dictionary). *For $t \in [b, c]$,*

$$X_k = \#^k(S^2 \times S^2),$$

and the fixed ascending spheres $G_i = S^2 \times \{p\}$ meet the descending red spheres $R_i(t)$ once at the endpoints. Isotopy extension and surgery give

$$(2.1.3) \quad \mathcal{C}_k: \pi_1 \left(\text{Emb} \left(\bigsqcup_{i=1}^k S^2, X_k \right) \right) \longrightarrow \pi_0(\text{Diff}(S^4)).$$

Lightbulb loops lie in $\ker \mathcal{C}_k$, and the punctured stable limit is onto [9]. Reordering every loop finger first produces its middle system $(G, R, \mathcal{F}, \mathcal{W})$.

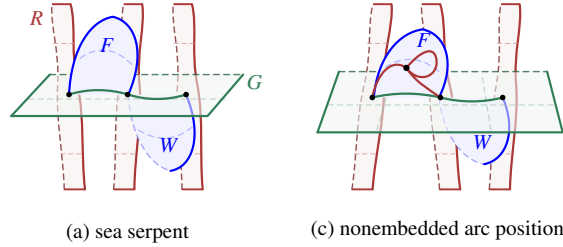


FIGURE 9. Two basic finger-Whitney systems.

Proposition 2.1.15 (Perturbed branch point, 3.34). *For $t \neq 0$, $z \mapsto z^2 + t\text{Re}(z)$ has a round critical circle and a t^2 -scaled three-cusped graphic, with folds elsewhere.*

Proof. Compute the Jacobian and restrict it to its zero set.

$$\begin{aligned} f_t(x, y) &= (x^2 - y^2 + tx, 2xy), & \det Df_t &= 4x^2 + 2tx + 4y^2, \\ \text{Crit}(f_t) &= \{(x + t/4)^2 + y^2 = t^2/16\}, & c_t(\theta) &= \frac{t}{4}(\cos \theta - 1, \sin \theta), \\ \gamma_t &= f_t c_t = \frac{t^2}{8}(\cos^2 \theta + \cos \theta - 2, (\cos \theta - 1)\sin \theta), \\ \gamma'_t &= \frac{t^2}{8}(2\cos \theta + 1)(-\sin \theta, \cos \theta - 1), \\ \gamma'_t = 0 &\iff \theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}, & |\det(\gamma'_t, \gamma''_t)| &= \frac{9t^4}{64} \neq 0. \end{aligned}$$

Thus precisely these three points are semicubical cusps. $\square$

Proposition 2.1.16 (Systems determine sphere paths, 3.35). *A system $(G, R, \mathcal{F}, \mathcal{W})$ gives a path from $R_{\mathcal{F}} \in \mathcal{LB}_k$, through R , to $R_{\mathcal{W}} \in \mathcal{LB}_k$, well defined relative to its endpoints. Closing it in $\mathcal{LB}_k$ gives a class of $\pi_1(\mathcal{E}_k)$ modulo left and right lightbulb loops.*

Proof. Run the finger movies backward and the Whitney movies forward.

$$\begin{aligned} P_{\mathcal{S}} &= \overline{\mathcal{F}} * \mathcal{W} : R_{\mathcal{F}} \rightsquigarrow R \rightsquigarrow R_{\mathcal{W}}, \\ |R_{\mathcal{F},i} \cap G_j| &= |R_{\mathcal{W},i} \cap G_j| = \delta_{ij}, \\ \pi_1(X_k) = 0 &\xrightarrow{[5]} \alpha_{\mathcal{F}}, \alpha_{\mathcal{W}} \subset \mathcal{LB}_k \text{ join the endpoints to } R^{\text{std}}, \\ L_{\mathcal{S}} &= \alpha_{\mathcal{F}} * P_{\mathcal{S}} * \alpha_{\mathcal{W}} \in \pi_1(\mathcal{E}_k), \\ L'_{\mathcal{S}} &= i_*(\ell_0) L_{\mathcal{S}} i_*(\ell_1), \quad i_* \pi_1(\mathcal{LB}_k) \subseteq \ker \mathcal{C}_k \xrightarrow{[9,6]} \mathcal{C}_k(L') = \mathcal{C}_k(L). \end{aligned}$$

Hence the ambiguity is exactly the double lightbulb coset. $\square$

Proposition 2.1.17 (Four model systems, 3.36, 3.37, 3.38, 3.39). *For Figures 9 and 9:*

- (1) *the sea serpent gives the constant path and $[\text{id}] \in \pi_0(\text{Diff}(S^4))$;*
- (2) *one interior F - W intersection gives $\mathcal{D} = 1$, hence no endpoint-relative contraction;*
- (3) *any 3-to-1 red-green slide crosses a tangency and leaves $\mathcal{LB}_k$;*
- (4) *the product ball gives two third Whitney disks, one disjoint from F , one disjoint from W , without excluding a disk disjoint from both.*

Proof. The pictured supports and intersection counts give all four conclusions.

$$\begin{aligned} \dot{F} \cap \dot{W} = \emptyset &\implies \overline{\mathcal{F}} * \mathcal{W} \simeq \overline{\mathcal{F}} * \mathcal{F} \simeq \text{const}, \\ |\dot{F} \cap \dot{W}| = 1 &\implies \mathcal{D} = I_1 + C = 1 \xrightarrow{2.2.8} [L_{\mathcal{S}}] \notin i_* \pi_1(\mathcal{LB}_k), \\ \#(R_s \cap G) : 3 \longrightarrow 1 &\implies \exists s_*, \quad R_{s_*} \not\cap G \implies R_{s_*} \notin \mathcal{LB}_k, \\ \partial U_F = \partial U_W = \alpha_R \cup \alpha_G, \quad \dot{U}_F \cap \dot{F} = \emptyset, \quad \dot{U}_W \cap \dot{W} = \emptyset. \end{aligned}$$

Thus the systems have the asserted classes and obstructions [6]. $\square$

2.2. Finger-Whitney systems and the five-move calculus.

Definition 2.2.1 (Embedding and lightbulb spaces). For $X_k = \#^k(S^2 \times S^2)$, put

$$R_i^{\text{std}} = \{g_0\} \times S^2, \quad G_i^{\text{std}} = S^2 \times \{r_0\}, \quad \mathcal{E}_k = \text{Emb}\left(\bigsqcup_i S^2, X_k\right), \quad * = R^{\text{std}},$$

$$\mathcal{LB}_k = \{R_1 \sqcup \cdots \sqcup R_k \in \mathcal{E}_k : R \cap G^{\text{std}}, |R_i \cap G_j^{\text{std}}| = \delta_{ij}\}.$$

Thus every red component has its designated geometric dual and misses the others [6, Definitions 0.1–0.2].

Definition 2.2.2 (Finger-Whitney systems). For $R \simeq R^{\text{std}}$ transverse to $G = G^{\text{std}}$, write

$$|R_i \cap G_j| = \delta_{ij} + 2n_{ij}.$$

A finger-Whitney system

$$\mathcal{S} = (G, R, \mathcal{F}, \mathcal{W}) \in \text{FW}_k$$

has internally disjoint complete framed pairings

$$\mathcal{F}_{ij} = f_{ij}^1 \sqcup \cdots \sqcup f_{ij}^{n_{ij}}, \quad \mathcal{W}_{ij} = w_{ij}^1 \sqcup \cdots \sqcup w_{ij}^{n_{ij}},$$

with $\mathcal{F} \cap \mathcal{W}$ allowed. Each disk has boundary $a_R \cup a_G$, clean interior, and the untwisted positive-normal finger germ or negative-normal Whitney germ [6, Definition 1.2].

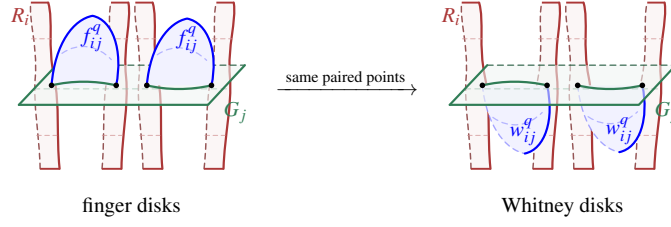


FIGURE 10. Finger and Whitney germs with the same paired intersection points.

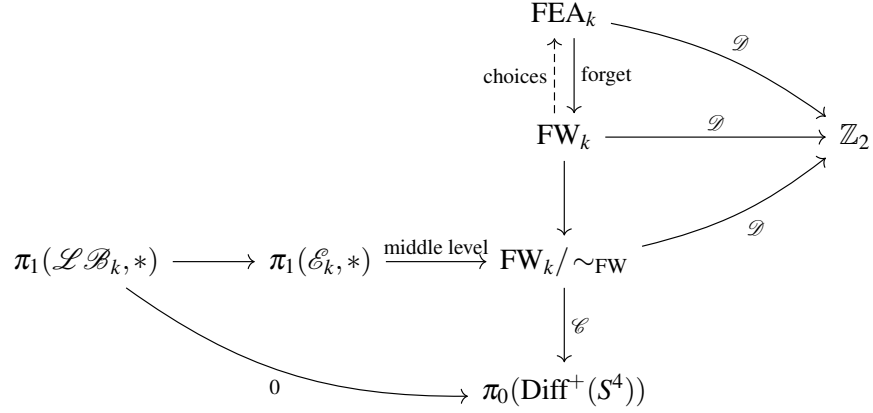


FIGURE 11. The reduction and detection diagram.

Definition 2.2.3 (Finger-first representatives). A generic based loop in $\mathcal{E}_k$ is finger first if all finger moves occur at $t = 1/4$ and all Whitney moves at $t = 3/4$; its middle system is

$$\mathcal{S}(R(t)) = (G^{\text{std}}, R(1/2), \mathcal{F}, \mathcal{W}).$$

Every class has such a representative [6, Definitions 0.6–0.8 and Construction 11.24].

Fact 2.2.4 (Stable loop-to-diffeomorphism dictionary). *The nested-eye construction induces*

$$\mathcal{C}_k: \pi_1(\mathcal{E}_k, *) \longrightarrow \pi_0(\text{Diff}(S^4)), \quad \text{im}(\pi_1(\mathcal{LB}_k, *) \rightarrow \pi_1(\mathcal{E}_k, *)) \subseteq \ker \mathcal{C}_k$$

[6, §0]. With compatible punctures $X_k^\dagger = X_k \setminus \{q_k\}$,

$$\varinjlim_k \mathcal{C}_k^\dagger: \varinjlim_k \pi_1(\mathcal{E}_k^\dagger, *) \twoheadrightarrow \pi_0(\text{Diff}(S^4))$$

[9, Theorem 1].

Definition 2.2.5 (Arc positions). For $\mathcal{S} = (G, R, \mathcal{F}, \mathcal{W})$:

- (1) $\mathcal{S} \in \text{IA}_k$ when the diagonal disk-boundary traces on each R_i, G_i are immersed arcs.
- (2) $\mathcal{S} \in \text{EA}_k$ when those arcs are embedded and ordered

$$f_{ii}^1, w_{ii}^1, \dots, f_{ii}^{n_{ii}}, w_{ii}^{n_{ii}}.$$

- (3) $\mathcal{S} \in \text{FEA}_k$ when it lies in EA_k and every off-diagonal trace is a disjoint embedded circle bounding a disjoint disk missing all other data.

These are the positions of [6, Definitions 0.9–0.10].

Definition 2.2.6 (The mod-two invariant). For $\mathcal{S} \in \text{FEA}_k$, counting disk interiors in $\mathbb{Z}_2$,

$$I_j(\mathcal{S}) = \sum_{p \leq q} |f_{jj}^p \cap w_{jj}^q|, \quad I = \sum_j I_j,$$

$$C_{ij} = |\mathcal{F}_{ij} \cap \mathcal{W}_{ji}|, \quad C = \sum_{i < j} C_{ij}, \quad \mathcal{D} = I + C$$

[6, Definition 0.10].

Theorem 2.2.7 (Reduction to full embedded arc position). *Every system reduces to FEA_k by untwisted isotopy, disk slides, boundary twisting, cross special sum-square moves, order-induced switching, and clasping; any two reductions have the same $\mathcal{D}$ [6, Corollary 9.4].*

Theorem 2.2.8 (Gabai–Gay–Hartman, . *GabaiGayHartman25*) For every $k \geq 1$,

$$\mathcal{D}: \pi_1(\mathcal{E}_k, *) \rightarrow \mathbb{Z}_2, \quad \pi_1(\mathcal{L}\mathcal{B}_k, *) \subseteq \ker \mathcal{D}.$$

A value-1 class exists for $k = 1$ and retains value 1 under every stabilization.

Definition 2.2.9 (Disk and sphere slides). A G -disk slide forms

$$K = d_2^+ \cup (\text{rectangle in } G) \cup d_2^-, \quad d_1 \mapsto d_1 \natural_G K;$$

an R -slide interchanges R, G . Closing $d^\pm$ over its move gives S_d , and a sphere slide is

$$d_2 \mapsto d_2 \natural_{S_{d_1}}$$

[6, Definitions 2.1 and 11.38].

Definition 2.2.10 (Birth, cubic, and saddle moves). (1) A birth creates two $R \cap G$ points and equal framed disks $f = w$, disjoint from the old data; death is its inverse. (2) An x^3 -move crosses a cusp whose point is free, paired by one old disk, or paired by an old finger disk and an old Whitney disk. (3) A saddle splits one old arc into a finger arc and a Whitney arc; finger-first reordering leaves one new pair. These are [6, Definitions 11.43, 11.47, and 11.51].

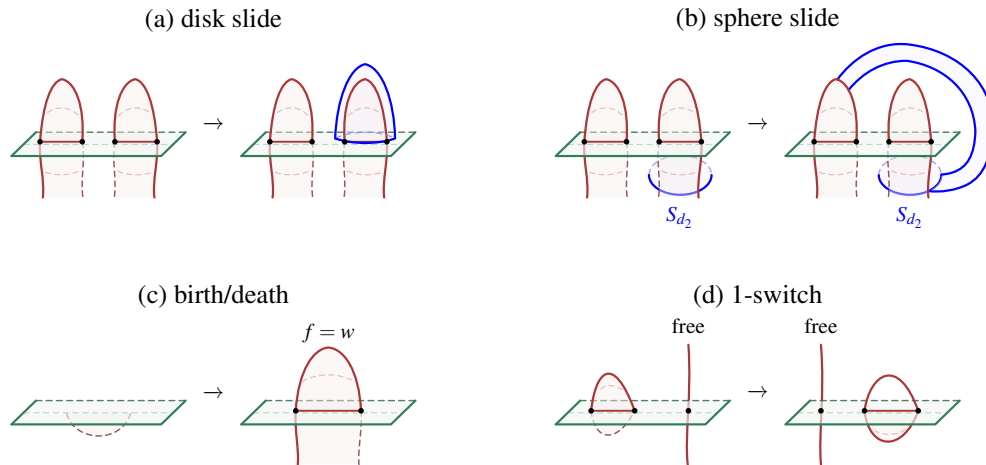


FIGURE 12. Four elementary finger–Whitney moves.



FIGURE 13. The special sum-square move.

Definition 2.2.11 (1-switch and special sum-square). A 1-switch replaces d by the complementary framed index-1 switch disk d^* . For a special sum-square move, choose same-germed d_1, d_2 and an opposite-germed Whitney disk D meeting both; resolve the two corner crossings with framed half-twisted bands and replace

$$\{d_1, d_2\} \mapsto \{d, D\}, \quad d = \text{resolved union of } d_1 \cup d_2 \cup D.$$

Exchange the germs for the finger version [6, Definitions 3.9 and 5.31].

Theorem 2.2.12 (The five-move theorem, . GabaiGayHartman25] *Based-homotopic finger-first loops have systems related by isotopy and finitely many*

$$\begin{array}{lll} R\text{- and } G\text{-disk slides,} & \text{finger- and Whitney-sphere slides,} & \\ \text{births/deaths,} & x^3\text{-moves,} & \text{saddle moves.} \end{array}$$

Conversely,

$$\mathcal{S} \sim_{\text{FW}} \mathcal{S}' \xrightarrow{[6, \S 11.7]} [R] = \ell_0[R']\ell_1, \quad \ell_0, \ell_1 \in \pi_1(\mathcal{LB}_k, *),$$

so equivalent systems have the same surgery class.

Proposition 2.2.13 (Version B). *An x^3_+ -move is a birth followed by one special sum-square or one 1-switch. A saddle is a birth followed by two moves, one on $\mathcal{F}$ and one on $\mathcal{W}$, each a special sum-square or a 1-switch; x^3_- reverses the sequence [6, Lemmas 12.12 and 12.14].*

Fact 2.2.14 (Trace ordering). *For a generic loop $R(t)$,*

$$\tilde{R} = \{(t, x) : x \in R(t)\}, \quad \tilde{G} = I \times G^{\text{std}}, \quad \tilde{R} \pitchfork \tilde{G}$$

is a 1-manifold whose time minima are finger moves and maxima are Whitney moves; reordering gives finger-first membranes [6, §11.2 and Construction 11.31]. For a generic homotopy $R(s, t)$,

graphic	move	graphic	move
index 0/1 fold	finger/Whitney	time min/max/saddle	birth/death/saddle
cuspid	x^3	normal crossing	reordering
arc meets boundary/interior	disk/sphere slide	brace	1-switch or sum-square

[6, §11.4 and Lemmas 11.35, 11.40].

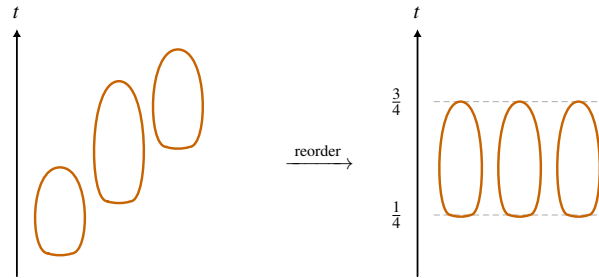


FIGURE 14. Reordering places every finger minimum at $t = \frac{1}{4}$ and every Whitney maximum at $t = \frac{3}{4}$.

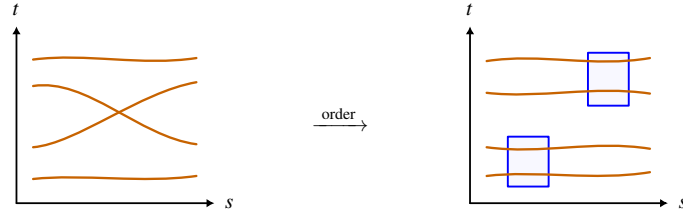


FIGURE 15. Ordering separates finger and Whitney folds into lower and upper bands; blue windows mark slide intervals.

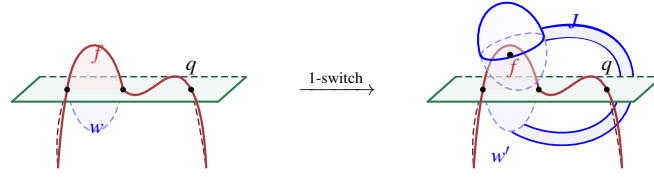


FIGURE 16. A 1-switch replaces w by $w' = w \# J$ and gives $|\mathring{f} \mathbin{\dot{\cap}} \mathring{w}'| = 1$.

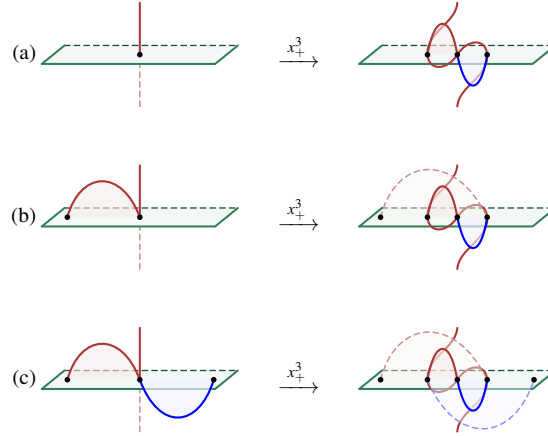


FIGURE 17. From top to bottom, x_+^3 extends zero, one, or two old corners and adds one disjoint finger–Whitney pair.

Proposition 2.2.15 (The key 1-switch, 3.40). *Figure 16 contains a 1-switch putting f, w' in embedded arc position with*

$$|\mathring{f} \mathbin{\dot{\cap}} \mathring{w}'| = 1.$$

Proof. Switch w across the free green point and use the red-finger meridian J .

$$\begin{aligned} w' &= w \# J, \\ \partial w' &= \text{complementary embedded switch arc} \xrightarrow{2.2.11} (f, w') \in \text{EA}_1, \\ A &= \text{tubing annulus}, \quad \mathring{w} \cap \mathring{f} = \emptyset, \quad A \cap f = \emptyset, \\ |J \mathbin{\dot{\cap}} f| &= 1, \\ |\mathring{f} \mathbin{\dot{\cap}} \mathring{w}'| &= |\mathring{f} \mathbin{\dot{\cap}} \mathring{w}| + |f \mathbin{\dot{\cap}} J| = 1. \end{aligned}$$

Thus the switch gives the unit example for $\mathcal{D}$ [6, Example 0.4 and Figure 1]. □

Proposition 2.2.16 (Why sphere slides occur, 3.41). *After one Whitney move, isotoping the second attaching arc back around the first changes its disk by a sphere slide.*

Proof. Track d_2 through the d_1 -move.

$$\begin{aligned} \gamma_2 &= \partial d_2 \cap (R \cup G), & B_1 &= 3\text{-ball swept out by the } d_1\text{-move,} \\ & & |\gamma_2 \cap B_1| &= 1, \\ d_2 &\xrightarrow{\text{transport}} \bar{d}_2, & \gamma_2 &\mapsto \bar{\gamma}_2, \\ \bar{\gamma}_2 &\xrightarrow{\text{isotopy across } B_1} \gamma_2 \implies d'_2 \simeq d_2 \# S_{d_1}, \\ d'_2 &\xleftrightarrow{2.2.9} \text{sphere slide.} \end{aligned}$$

Thus interchanging the Whitney operations forces the move [6, Definition 11.38, Remark 11.39, and Lemma 11.40]. $\square$

Proposition 2.2.17 (The three cubic flavors, 3.42). *All three x^3 -flavors in Figure 17 are generated by the other moves.*

Proof. Birth a framed pair (f_0, w_0) disjoint from the old data.

$$\begin{aligned} B &= \text{birth}, & S &= \text{special sum-square}, & Q &= 1\text{-switch}, \\ & & \text{two old disks : } x_+^3 &= SB, \\ & & \text{one old disk : } x_+^3 &= QB, \\ \text{free point : } x_+^3 &= QB \xrightarrow{[6, \text{Lemma 12.12}]} x_+^3 \in \langle B, S, Q \rangle, \\ x_-^3 &= (x_+^3)^{-1} \xrightarrow{2.2.13} \text{reverse the Version B sequence.} \end{aligned}$$

Thus the remaining moves generate every cubic flavor. $\square$

Part 2. Exercises**3. EXERCISES****Exercise 3.1** (1.1.4). Show

$$\pi_0(\text{Homeo}_\partial(B^n)) = 0.$$

Exercise 3.2 (1.1.5). Show

$$\pi_0(\text{Diff}_\partial(B^1)) = 0.$$

Explain why the proof fails for $n > 1$.**Exercise 3.3** (1.1.6). Show

$$\pi_0(\text{Diff}_\partial(B^n)) = \pi_0(\text{Diff}(S^n)).$$

Exercise 3.4 (1.1.7). Show

$$\pi_0(\text{Diff}(B^n)) \quad \text{and} \quad \pi_0(\text{Diff}(S^n)) \quad \text{are abelian.}$$

Exercise 3.5 (1.1.8). Show

$$Y_\phi \cong_{\text{bun}} Y_{\phi'} \implies (\exists h \in \text{Diff}(X)) \phi' \simeq h\phi h^{-1}.$$

If the bundle isomorphism fixes $\{0\} \times X$, show

$$\phi' \simeq \phi.$$

Exercise 3.6 (1.1.9). Show

$$\partial Z = Y, \quad f: Y \rightarrow S^1 \implies (\exists F: Z \rightarrow D^2) F|_Y = f,$$

and determine $\text{Sing}(F)$.**Exercise 3.7** (1.1.10). Show

$$Y_\phi \rightarrow S^1 \text{ extends to a nonsingular } Z \rightarrow D^2 \implies \phi \simeq \text{id}.$$

Exercise 3.8 (1.1.11). For

$$p(z_0, \dots, z_m) = z_0^2 + \dots + z_m^2,$$

show

$$\text{Crit}(p) = \{0\}, \quad \text{rk}(dp_0) = 0.$$

Exercise 3.9 (1.1.13). Let

$$p: \mathbb{C}^{m+1} \longrightarrow \mathbb{C}, \quad p(z_0, \dots, z_m) = z_0^2 + \dots + z_m^2, \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^{m+1}.$$

Compute its monodromy.

(1) Show

$$p^{-1}(1) = \{\mathbf{x} + i\mathbf{y} : |\mathbf{x}|^2 = 1 + |\mathbf{y}|^2, \mathbf{x} \cdot \mathbf{y} = 0\}.$$

(2) Show

$$TS^m = \{(\mathbf{x}, \mathbf{y}) : |\mathbf{x}| = 1, \mathbf{x} \cdot \mathbf{y} = 0\}.$$

(3) Show

$$(\mathbf{x}, \mathbf{y}) \longmapsto \sqrt{1 + |\mathbf{y}|^2} \mathbf{x} + i\mathbf{y}$$

is a diffeomorphism

$$TS^m \longrightarrow p^{-1}(1).$$

(4) For

$$\phi_r(\mathbf{x}, \mathbf{y}) = \sqrt{r + |\mathbf{y}|^2} \mathbf{x} + i\mathbf{y},$$

show

$$r > 0 \implies \phi_r: TS^m \xrightarrow{\cong} p^{-1}(r), \quad r = 0 \implies \phi_0 \text{ collapses the zero-section.}$$

(5) For $r > 0$, show

$$(\mathbf{x}, \mathbf{y}) \longmapsto e^{i\theta/2} \phi_r(\mathbf{x}, \mathbf{y})$$

is a diffeomorphism

$$TS^m \longrightarrow p^{-1}(re^{i\theta}).$$

(6) For $\mathbf{y} \neq 0$, let

$$R_\rho(\mathbf{x}, \mathbf{y}) = \left(\cos \rho \mathbf{x} + \sin \rho \frac{\mathbf{y}}{|\mathbf{y}|}, -\sin \rho |\mathbf{y}| \mathbf{x} + \cos \rho \mathbf{y} \right).$$

Show

$$R_\rho(\mathbf{x}, \mathbf{y}) = \left(\Re \left(e^{-i\rho} \left(\mathbf{x} + i \frac{\mathbf{y}}{|\mathbf{y}|} \right) \right), \Im \left(e^{-i\rho} (|\mathbf{y}| \mathbf{x} + i \mathbf{y}) \right) \right),$$

and extend R_0 by the identity at $\mathbf{y} = 0$.

(7) Show

$$l(0) = 0 \implies R_l(\mathbf{x}, \mathbf{y}) := R_{l(|\mathbf{y}|)}(\mathbf{x}, \mathbf{y}) \text{ is a diffeomorphism.}$$

Fix

$$\tau: [0, \infty) \rightarrow [0, 1], \quad \tau|_{[0, \varepsilon]} = 0, \quad \tau|_{[1-\varepsilon, \infty)} = 1,$$

with τ increasing between these intervals, and define

$$\psi(re^{i\theta}, \mathbf{x}, \mathbf{y}) = e^{i\theta/2} \phi_r(R_{(\theta/2)\tau(|\mathbf{y}|)}(\mathbf{x}, \mathbf{y})).$$

(8) For $|\mathbf{y}| \geq 1 - \varepsilon$, show

$$\psi(0e^{i\theta}, \mathbf{x}, \mathbf{y})$$

is independent of θ .

(9) For $|\mathbf{y}| \geq 1 - \varepsilon$, show that the values at $\theta = 0, 2\pi$ agree and induce

$$\Psi: \mathbb{C} \times (TS^m \setminus \{|\mathbf{y}| < 1 - \varepsilon\}) \longrightarrow \mathbb{C}^{m+1}.$$

(10) Show

$$p\left(\psi(re^{i\theta}, \mathbf{x}, \mathbf{y})\right) = re^{i\theta},$$

and deduce that ψ trivializes p away from

$$r = 0, \quad \mathbf{y} = 0.$$

(11) Show

$$\psi(r, -R_{\pi\tau(|\mathbf{y}|)}(\mathbf{x}, \mathbf{y})) = \psi(re^{2\pi i}, \mathbf{x}, \mathbf{y}).$$

(12) Deduce

$$\text{Mon}(p) = R_{-\pi\tau}.$$

Exercise 3.10 (1.1.15). Show that the Dehn-twist class depends only on the isotopy class of its sphere.

Exercise 3.11 (1.1.16). For $m = 1$, show that $R_{-\pi\tau}$ is the usual right-handed Dehn twist.

Exercise 3.12 (1.1.17). Show

$$[R_{-\pi\tau}] \neq [\text{id}] \quad \text{in} \quad \pi_0(\text{Diff}_\partial(DTS^m)).$$

It suffices to treat dimension 2.

Exercise 3.13 (1.1.18). For $m = 2$, show that the Dehn twist has order 2 in the smooth mapping class group. Determine its order for $m > 2$.

Exercise 3.14 (1.2.1). Show

$$S^3 = \partial B^4 \subset X^4 \implies \tau_{S^3} \simeq \text{id}_X.$$

Exercise 3.15 (1.2.2). Show

$$S^1 \times \Sigma = \partial(D^2 \times \Sigma) \implies \tau_{S^1 \times \Sigma} \simeq \text{id}.$$

Exercise 3.16 (1.2.5). For a Montesinos twin $W = (R, S)$, show

$$\partial \nu W \cong S_\lambda^1 \times S_R^1 \times S_S^1.$$

Exercise 3.17 (1.2.6). Show that orientations may be chosen so that

$$\tau_{(R,S)}^{-1} = \tau_{(S,R)} = \tau_{(-R,S)} = \tau_{(R,-S)}.$$

Exercise 3.18 (1.2.7). For the trivial twin $W_0 = (R_0, S_0)$, show

$$\tau_{W_0} \simeq \text{id}.$$

Exercise 3.19 (1.2.10). Let $W = (R, S) \subset X^4$ and

$$X^* = (X \setminus \nu R) \cup_{S^2 \times S^1} (B^3 \times S^1).$$

(1) Show that S becomes a torus

$$T \subset X^*.$$

(2) Show

$$T = \bigcup_{t \in S^1} \gamma_t, \quad \gamma_t \subset X^*, \quad \gamma_0 = \gamma_1 = \mu_R,$$

where $\mu_R \subset S$ bounds a small disk about one point of $R \cap S$.

(3) Show that surgery on

$$\mu_R \subset X^*$$

recovers X .

(4) Show that circle pushing along $\{\gamma_t\}$, followed by surgery on μ_R and extension over X , induces τ_W .

Exercise 3.20 (1.2.11). For a spun-knot twin $W = (R, S) \subset S^4$, show

$$\tau_W \simeq \text{id}.$$

Exercise 3.21 (1.2.12). Show that the two definitions of a barbell map agree.

Exercise 3.22 (1.2.16). Show

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^3 \implies f \text{ is not stable}.$$

Exercise 3.23 (1.2.17). Show

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2 \implies f \text{ is stable}.$$

Exercise 3.24 (2.1.3). Show that the cusp-model singular locus is a smoothly embedded 1-manifold.

Exercise 3.25 (2.1.5). For a Morse 2-function $F: Z \rightarrow [0, 1]^2$, show

$$\text{the graphic has no vertical tangencies} \implies Z \cong [0, 1] \times Y, \quad Y = F^{-1}(\{0\} \times [0, 1]).$$

Exercise 3.26 (2.1.3). Show

$$F: Z \rightarrow D^2 \text{ a Morse 2-function} \implies \text{Sing}(F) \subset Z \text{ is a smoothly embedded 1-manifold}.$$

Exercise 3.27 (2.1.5). Let $F: Z \rightarrow [0, 1]^2$ have no vertical tangencies and no singularities over $\partial[0, 1]^2$. For

$$Y = F^{-1}(\{0\} \times [0, 1]), \quad X = F^{-1}(0, 0),$$

show

$$Y \cong [0, 1] \times X, \quad Z \cong [0, 1] \times Y \cong [0, 1]^2 \times X, \\ F(t, y, x) = (t, f_t(y, x)),$$

where f_t is Morse except at finitely many birth–death times.

Exercise 3.28 (2.1.5). Show that the boundary monodromy satisfies

$$\phi \simeq_{\text{pseudo}} \text{id}.$$

Exercise 3.29 (2.1.7). Construct a 1-parameter family of gradient-like vector fields for functions which are Morse except at finitely many birth–death times.

Exercise 3.30 (2.1.7). Show that immediately below a birth the new descending manifold is a ball which becomes the lower-index descending sphere and the higher-index punctured descending sphere.

Exercise 3.31 (2.1.7). Let X be simply connected and

$$I = [1/4, 3/4], \quad J = [1/4, 1/4 + \varepsilon] \cup [3/4 - \varepsilon, 3/4].$$

For

$$\alpha_t: S^0 \hookrightarrow X \quad (t \in I), \quad \beta_t: B^1 \hookrightarrow X \quad (t \in J), \quad \beta_t|_{S^0} = \alpha_t,$$

show that some $\psi_t \in \text{Diff}(X)$, $t \in [0, 1]$, satisfies

$$\psi_0 = \psi_1 = \text{id}, \quad \psi_t \circ \alpha_t = \alpha_0 \quad (t \in I), \quad \psi_t \circ \beta_t = \beta_0 \quad (t \in J).$$

Exercise 3.32 (2.1.9). Show

$$S^1_{\text{dotted}} \subset Y^3 \longleftrightarrow \text{a 4-dimensional 1-handle on } Y^3 \times [0, 1], \\ S^2_{\text{dotted}} \subset X^4 \longleftrightarrow \text{a 5-dimensional 1-handle on } X^4 \times [0, 1].$$

Exercise 3.33 (2.1.10). Show that circle pushing equals Montesinos twisting and that the 1–2 subgroup is generated by half-unknotted Montesinos twists.

Exercise 3.34 (2.1.15). For

$$f_t(z) = z^2 + t\Re(z),$$

determine its singular locus and graphic, classify its folds and cusps, and draw the graphic.

Exercise 3.35 (2.1.16). (1) Show that (G, R, F, W) determines, up to homotopy rel endpoints in LB, a path of red spheres with endpoints in LB.

(2) Use Gabai’s lightbulb theorem to obtain an element of

$$\pi_1 \left(\text{Emb} \left(\bigsqcup_{i=1}^k S^2, \#^k(S^2 \times S^2) \right) \right).$$

(3) Determine its indeterminacy.

Exercise 3.36 (2.1.17). Using the upper-left system in Figure 9, show

disjoint finger- and Whitney-disk interiors $\implies$ the sphere path is null-homotopic $\implies [\text{id}] \in \pi_0(\text{Diff}(S^4))$.

Exercise 3.37 (2.1.17). Explain why the preceding argument fails for the upper-right system in Figure 9, where a finger disk meets a Whitney disk once.

Exercise 3.38 (2.1.17). Identify the obstruction to sliding the upper-right system in Figure 9 off the green sphere while the red sphere path remains in lightbulb position.

Exercise 3.39 (2.1.17). In Figure 9, find a third Whitney disk pairing the unpaired point with another point and missing one indicated F - or W -disk. Show that it cannot miss both.

Exercise 3.40 (2.2.15). Interpret Figure 16 as the time-1/2 level of a finger-first loop of red spheres, and find a 1-switch producing embedded arc position with

$$|\mathring{f} \cap \mathring{w}'| = 1.$$

Exercise 3.41 (2.2.16). Verify the sphere-slide move by one Whitney move followed by an isotopy of the remaining Whitney disk.

Exercise 3.42 (2.2.17). Using Figure 17, realize all three x^3 -moves as combinations of the other moves.

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